ECE501 - DIGITAL IMAGE PROCESSING

SAR-Optical Feature Fusion for Crop-Type Classification
REPORT - WEEK 4

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GitHub repository:
https://github.com/srushtibeng-svg/DIP_2026_Project19-SOFF_Group20.git

Assigned Data: Germany + Uruguay (Group A), YieldSAT Sentinel-2 + crop labels + yield masks; Sentinel-1 SAR acquired by the group via Google Earth Engine.

1. Project Overview

The project combines optical (Sentinel-2) and SAR (Sentinel-1) features for crop-type classification. Optical imagery can be affected by cloud cover, while Sentinel-1 SAR provides complementary observations in cloudy conditions. The current study uses YieldSAT sample data for Germany and Uruguay.

2. Work Completed This Week

- **Dataset inspection:** Continued examining the Germany and Uruguay sample CSV files, including crop labels, dates, field identifiers and optical-band columns.

- **Germany optical analysis:** Reviewed the preliminary NDVI analysis for 50 Germany samples using NDVI = (B08 − B04) / (B08 + B04). All 50 samples produced valid NDVI values.
- **Preliminary NDVI statistics:** For the 50-sample Germany subset, NDVI ranged from 0.3129 to 0.8780, with a mean of 0.5959 and a standard deviation of 0.1460.
- **Optical feature extraction code:** Wrote `optical_features.py` and `run_optical_extraction.py`, which compute NDVI, NDRE, NDMI and EVI from the five Sentinel-2 bands (B02, B04, B05, B08, B11) and summarise each index per field by mean, standard deviation and maximum. The code passes unit tests on synthetic data and has not yet been run on the full Germany data.
- **Plotting and Sentinel-1 scripts:** Prepared scripts for index time-series plots, rapeseed-vs-wheat feature boxplots and a Google Earth Engine Sentinel-1 (VV, VH) time-series export. These have not yet been run on the real data.
- **Spatial-information investigation:** Investigated the available dataset metadata and geometry information needed to associate sample fields with Sentinel-1 observations.

3. Preliminary Results

Measure	Germany sample (n = 50)	Uruguay (n = 50)
Valid NDVI values	50 / 50	50 / 50
Minimum NDVI	0.3129	0.2947
Mean NDVI	0.5959	0.6695
Standard deviation	0.1460	0.1977
Maximum NDVI	0.8780	0.9186

These values describe only the 50-sample Germany subset. They are preliminary optical-data results, not final crop-classification performance. A verified paired-band NDVI result for Uruguay is not included in this report. The Germany sample contains 25 rapeseed and 25 wheat rows

(mean NDVI 0.5755 and 0.6164 respectively). The two crops are sampled on different dates, so this difference is indicative only.

4. Challenges and Current Limitations

- Reliable field-level geographic coordinates or valid field geometries have not yet been confirmed for the current sample records.
- The available sample geometry information is insufficient to match each field to a Sentinel-1 observation with confidence.
- SAR extraction, SAR–optical feature fusion and final crop-classification evaluation are therefore still pending.
- The NDVI statistics come from a small sample of 50 Germany records and should not be generalised to the full dataset.

5. Plan for Next Week

1. Check the original YieldSAT documentation and metadata for a reliable source of field-level spatial information, and ask the course staff if none is found.
2. Run the optical extraction code and plotting scripts on the Germany sample and save the feature table and plots to the repository.
3. Validate the spatial information, then extract Sentinel-1 VV and VH features for the Germany and Uruguay samples.
4. If reliable matching becomes possible, build a field/date-based table linking optical and SAR features.
5. Report fusion or model-accuracy results only after they have been experimentally verified.

6. Summary

In Week 4 we reviewed the sample datasets, documented the preliminary Germany NDVI results, wrote the optical feature extraction code and investigated the spatial information needed for SAR matching. The main dependency remains reliable field-level geolocation; once it is resolved, we will proceed to Sentinel-1 feature extraction and fusion.